

Saam Haghighat-Grami

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Roswell, GA

Interests – Autonomous, Mobile, Collaborative Robotics

Graduating December 2026

LinkedIn – saam-haghighat-grami

GitHub – Saam-Grami

Portfolio – saam-grami.github.io

Active Military Secret Security Clearance

EDUCATION

- Kennesaw State University** *January 2022 – May 2022, January 2024 – Present*
Mechatronics Engineering, B.S. – Graduation December 2026 CGPA: 4.00
- Marine Corps Communication Electronics School** *August 2022 – October 2023*
Ground Electronics Transmission Systems Maintainer CGPA: 4.00

TECHNICAL SKILLS

Autonomy & Robotics: ROS 2, Gazebo, Nav2, ArduPilot, PX4, Mission Planner, UAV/UGV integration, RTAB Map
Embedded & Hardware: ESP32, Arduino, Raspberry Pi 5, Jetson Orin Nano, OpenMV, LiDAR, RGB-D cameras, IMUs, ad hoc networking, Control Theory
Languages & Tools: C++, Python, MATLAB, LaTeX, Linux/WSL2, Git, VS Code, SolidWorks, Onshape
Perception & ML: PyTorch, TensorFlow, YOLO, CNNs, LSTMs, OpenCV, ORB-SLAM3, sensor fusion
Manipulators & Automation: Denavit-Hartenberg parameters, forward and inverse kinematics, trajectory planning, Ladder Logic, TIA Portal, PLCs, HMI development

Portfolio — saam-grami.github.io — Full publication list, in-progress projects (multi-drone swarm sim, GPS-denied UAV-UGV localization), and additional technical detail.

RESEARCH & ENGINEERING EXPERIENCE

- MEAP – Mass Estimation Autonomous Pathfinding for Search and Rescue** *November 2025 - Present*
Robotics Software Architect & Project Coordinator – AgriSys Lab, Dr. Muhammad Hassan Tanveer
– UAV-UGV collaborative search-and-rescue system where the UAV stitches aerial maps and segments objects below while the UGV navigates via SLAM, fusing its sensor data with the UAV's to layer maps and estimate each object's volume and mass, deciding whether it can be moved to shorten the path to the objective.
– Leading a 5-person team from planning through execution; built the UAV autonomy stack (flight motion framework, aerial map stitching), validated on a 2 m² area and scaling toward 25 m². Also leading the ML pipeline that estimates obstacle mass from fused SLAM point clouds, targeting 85% mission success on obstacles up to ~0.5 kg.
- DARB – Drone Arm Robotic Backpack (Personal Project)** *May 2025 - Present*
Sole Designer and Builder
– Backpack-mounted robotic arm that autonomously captures a drone mid-flight, stows it, and relauches it, aiming for a 30-second capture-to-relaunch cycle from stable hover.
– Built and flight-tested a working end-to-end system – custom airframe, ArduPilot-based capture arm (Matek H743, NEMA 23 steppers), and an AprilTag vision-alignment pipeline – validated across repeated live test flights; now tuning the vision pipeline for more consistent lock-on.
- Trajectory Forecasting for Floating Debris in Waterways** *June 2026 - August 2026*
Funded Undergraduate Researcher – Summer Undergraduate Research Program, Dr. Matt Marshall
– Real-time vision system that detects, tracks, and forecasts floating river debris trajectories from a fixed riverbank camera to direct a collection platform toward interception.
– Combines dense optical-flow surface velocimetry with object motion/slip in a CNN + LSTM forecaster; a Gaussian head yields calibrated multi-step forecasts over YOLO26 detections, with identity reconciled across lost tracks.
- SoilBus – Soil Monitoring Probe Network** *August 2025 - December 2025*
Lead Software Developer – AgriSys Lab, Dr. Muhammad Hassan Tanveer
– Hub-less soil-monitoring mesh network where probes auto-discover neighbors and any single probe can offload the whole network's data.
– Reached full mesh connectivity in 15s, with each node offloading a day's sensor data in ~5s; BFS-based routing validated on 5 physical nodes, holding to 26 simulated before degrading.

PUBLICATIONS

Author on 3 works in autonomous systems research – 1 published (IEEE SoutheastCon 2026, IEEE Xplore), 2 first-author manuscripts in preparation for peer reviewed computer vision venue and IROS 2027. Full citations and DOIs at portfolio above.

INDUSTRY & MILITARY EXPERIENCE

- EOSYS Industrial and Manufacturing System Integrator** *May 2025 – August 2025*
COOP Marietta GA
– Built an HMI display generator that produced 40+ P&ID screens for plant operators.
– Reviewed and modified PLC programs across two production shops plus the PLC trainer system.
- United States Marine Corps Reserve** *August 2022 - Present*
Ground Radio Repair Specialist Marietta GA
– Troubleshoot, maintain, and operate USMC radio systems in time-sensitive conditions; active Secret clearance.
– Lead a squad of 8 Marines in the 4th Combat Readiness Regiment, responsible for training, accountability, and readiness.